

Linear algebra – 9

Matrices over $\mathbb{Z}$ and over a finite field

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Theoretical minimum.

Def. $GL_n(\mathbb{Z})$ consists of the integer matrices with $\det = \pm 1$; a *lattice* in $\mathbb{R}^n$ is $\Lambda = AZ^n$ with A invertible, and $[\mathbb{Z}^n : \Lambda] = |\det A|$ when $\Lambda \subseteq \mathbb{Z}^n$.

Theorem 1 (Smith normal form). Every $A \in M_{m,n}(\mathbb{Z})$ can be written as $A = U \operatorname{diag}(d_1, \dots, d_r) V$ with U, V unimodular and $d_1 \mid d_2 \mid \dots \mid d_r$; the d_i are determined by $d_1 \cdots d_k = \gcd$ of all $k \times k$ minors.

Theorem 2 (counting matrices). $|GL_n(\mathbb{F}_q)| = \prod_{k=0}^{n-1} (q^n - q^k)$, and the number of k -dimensional subspaces of $\mathbb{F}_q^n$ is the Gaussian binomial coefficient $\binom{n}{k}_q$.

Fact 1. A polynomial identity between determinants that holds over $\mathbb{Z}$ holds over every commutative ring; conversely, reduction modulo a prime is the cheapest way to prove that an integer determinant is nonzero.

1. Prove the existence and uniqueness of the Smith normal form, and deduce the structure theorem for finitely generated abelian groups as well as the formula $[\mathbb{Z}^n : \Lambda] = |\det A|$.
2. (IMC) Let A be an $n \times n$ matrix with integer entries and let $b_1, \dots, b_n$ be integers with $\det A = b_1 \cdots b_n$. Prove that there exist integer matrices $B_1, \dots, B_n$ with $A = B_1 \cdots B_n$ and $\det B_i = b_i$ for all i .
3. (Putnam 2021) Call an $n \times n$ integer matrix $A = (a_{ij})$ *very odd* if for every non-empty $S \subseteq \{1, \dots, n\}$ the submatrix $(a_{ij})_{i,j \in S}$ has odd determinant. Prove that if A is very odd, then so is A^k for every $k \geq 1$.
4. (IMC) Let A be a symmetric $m \times m$ matrix over the two-element field all of whose diagonal entries are zero. Prove that for every positive integer n each column of A^n has a zero entry.

5. (ФКН, осень 2023) All entries of a 3×3 matrix are squares of odd numbers. Prove that its determinant is divisible by 64.